Assignment 3: Malware Feature Clustering

Item	Requirement
Submission file	Assignment3-LastName.ipynb
Dataset	UCI Malware static and dynamic features VxHeaven and VirusTotal
Required methods	K-Means, Hierarchical Clustering, DBSCAN, Silhouette Score
AI use	Allowed only for the AI-assisted interpretation part with documentation
Total points	100

Introduction

In Lab 3, you completed a guided malware-feature clustering workflow using the CIC-Evasive-PDFMal2022 dataset. In this assignment, you will complete a more independent clustering analysis using static and dynamic malware feature data from benign Windows files, VxHeaven malware files, and VirusTotal malware files.

You will combine the dataset files, clean and scale numeric features, compare K-Means, hierarchical clustering, and DBSCAN, evaluate cluster quality, inspect cluster profiles, and explain the results from a malware-analysis perspective.

This assignment is based on unsupervised learning approach. Labels are used only after clustering to help interpret the results. Do not use source labels or malware labels as clustering features.

Dataset

Required File	Description	Label to Add
staDynBenignLab.csv	Static and dynamic features from benign Windows files	source_label = benign; binary_label = benign
staDynVxHeaven2698Lab.csv	Static and dynamic features from VxHeaven malware files	source_label = malware_vxheaven; binary_label = malware
staDynVt2955Lab.csv	Static and dynamic features from VirusTotal malware files	source_label = malware_virustotal; binary_label = malware

Important dataset rules:

- Use only the CSV feature files.
- Create source_label and binary_label after loading the files.
- Do not use source_label, binary_label, file names, hashes, or identifiers as clustering features.
- Use labels only after clustering for interpretation, comparison tables, and discussion.
- Use a controlled analysis subset if needed so the notebook runs in a reasonable time.

AI Use Policy

AI use is allowed only for the designated AI-assisted interpretation and documentation part of this assignment. You may not submit an AI-generated notebook that you do not understand. You are responsible for testing, correcting, and explaining everything you submit.

Allowed AI use:

- Explaining code errors or debugging messages
- Explaining clustering concepts such as linkage, eps, MinPts, silhouette score, or cluster profiles
- Helping interpret aggregate cluster-profile tables that you generated
- Helping rewrite your own interpretation for clarity
- Suggesting questions to ask about unusual clusters or DBSCAN noise points

Not allowed:

- Submitting AI-generated work that you do not understand
- Submitting code that you cannot explain
- Fabricating results, plots, metrics, prompts, or AI outputs
- Using labels as clustering features and presenting the result as unsupervised learning
- Pasting full datasets, raw malware samples, file hashes, or sensitive identifiers into an AI tool
- Copying another student's notebook or report

At the end of your notebook, include an AI Use Statement and a prompt documentation table. If you did not use AI tools, state that directly. However, Part 10 requires AI-assisted analysis documentation for full credit.

Required Packages

Use the following packages. You may import additional packages if needed.

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.cluster import KMeans, AgglomerativeClustering, DBSCAN
from sklearn.metrics import silhouette_score, adjusted_rand_score,
normalized_mutual_info_score
from sklearn.neighbors import NearestNeighbors
from scipy.cluster.hierarchy import linkage, dendrogram
```

Part 1: Dataset Loading and Combination

Load the three required CSV files and combine them into one dataframe. Add source_label and binary_label before combining.

1. Load staDynBenignLab.csv, staDynVxHeaven2698Lab.csv, and staDynVt2955Lab.csv.
2. Display the shape of each file before combining.
3. Add source_label and binary_label columns to each dataframe.
4. Combine the three dataframes into one dataframe.
5. Display the combined shape, column count, and label distributions.
6. Briefly explain why labels must not be used as clustering features.

Required outputs:

- First five rows of the combined dataframe
- Shape of each original file and the combined dataframe
- source_label and binary_label distribution tables
- Short explanation of the unsupervised learning rule

Part 2: Data Cleaning and Feature Selection

Prepare a numeric feature matrix for clustering. Remove label columns and any identifier-like columns.

7. Identify numeric and nonnumeric columns.
8. Remove source_label, binary_label, file names, hashes, and identifiers from the feature matrix.
9. Handle missing values and infinite values.
10. Drop constant or near-constant columns.
11. Select a reasonable feature set for clustering. You may use all cleaned numeric features or a selected subset, but your choice must be explained.
12. Create a table explaining the feature groups or selected features.

Feature or Feature Group	Reason for Keeping It	Possible Malware-Analysis Meaning

Required outputs:

- Missing value summary before and after cleaning
- List of excluded columns
- Final number of selected features
- Feature selection table

Part 3: Scaling and Exploratory Analysis

Scale the selected numeric features and create visual summaries before clustering.

13. Scale numeric features using StandardScaler.
14. Create at least four histograms or boxplots for selected features.
15. Create one correlation heatmap using a manageable set of selected features.
16. Create a 2D PCA visualization colored by source_label or binary_label for interpretation only.
17. Briefly explain visible patterns, outliers, skewed features, and correlated features.

Required outputs:

- Scaled feature matrix shape
- At least four feature distribution plots
- One correlation heatmap
- One PCA scatter plot
- Short exploratory interpretation

Part 4: K-Means Clustering and Choosing K

Run K-Means for several K values and use cluster-quality evidence to choose a reasonable K.

18. Run K-Means for K values from 2 to 10.
19. Use random_state=40103.
20. Record inertia and silhouette score for each K.
21. Create an elbow plot and a silhouette score plot.
22. Choose one K and explain your choice.
23. Visualize the selected K-Means clusters using PCA coordinates.

K	Inertia	Silhouette Score	Comment
2			
3			
4			
5			

Required outputs:

- Table of K, inertia, and silhouette score
- Elbow plot
- Silhouette score by K plot
- PCA scatter plot colored by K-Means cluster
- Short explanation of selected K

Part 5: Hierarchical Clustering and Linkage Comparison

Use hierarchical clustering to inspect relationships among samples. Because dendrograms can be expensive, use a controlled sample and report the sample size.

24. Create a sample for the dendrogram. Use no more than 300 records unless your computer can handle more.
25. Build and display a dendrogram using Ward linkage.
26. Run agglomerative clustering using single, complete, average, and Ward linkage.
27. Use the same K value or a small set of K values for fair comparison.
28. Compare silhouette scores across linkage methods.
29. Explain what each linkage method means and how it changes cluster behavior.

Linkage Method	K	Silhouette Score	Interpretation
Single			Nearest cross-cluster pair
Complete			Farthest cross-cluster pair
Average			Average cross-cluster distance
Ward			Compact clusters with lower within-cluster spread

Required outputs:

- Dendrogram with reported sample size
- Linkage comparison table
- PCA scatter plot for at least one hierarchical clustering result
- Short explanation of single, complete, average, and Ward linkage

Part 6: DBSCAN Parameter Analysis

Use DBSCAN to find dense groups and identify unusual samples as noise.

30. Create a k-distance plot to guide eps selection.
31. Try several eps values and at least three MinPts values.
32. For each setting, report the number of clusters, number of noise points, noise rate, and silhouette score when valid.
33. Choose one DBSCAN setting and explain why it is reasonable.
34. Visualize the selected DBSCAN result using PCA coordinates.
35. Explain what core points, border points, and noise points mean in this experiment.

eps	MinPts	Clusters	Noise Rate	Silhouette Score	Comment

Required outputs:

- k-distance plot
- DBSCAN parameter sweep table
- Selected eps and MinPts values
- PCA scatter plot colored by DBSCAN cluster, including noise if present
- Short explanation of DBSCAN point types and parameter effects

Part 7: Cluster Evaluation and Label Comparison

Compare the clustering results using internal cluster-quality metrics and post-clustering label comparison. Labels must not be used during clustering.

36. Report silhouette score for the selected K-Means result.
37. Report silhouette score for the selected hierarchical clustering result.
38. Report silhouette score for DBSCAN if at least two non-noise clusters exist.
39. Use crosstab tables to compare clusters with source_label and binary_label after clustering.
40. Optionally compute adjusted Rand index or normalized mutual information as external comparison metrics.
41. Explain why high label alignment does not automatically mean the clusters are useful.

Method	Parameters	Number of Clusters	Silhouette Score	Label Alignment Comment
K-Means				
Hierarchical				
DBSCAN				

Required outputs:

- Internal metric comparison table
- Cluster versus source_label crosstab
- Cluster versus binary_label crosstab
- Short interpretation of what the labels reveal after clustering

Part 8: Cluster Profile Analysis

Cluster IDs do not explain themselves. Inspect feature summaries to understand what each cluster represents.

42. Compute the mean or median feature profile for each selected clustering method.
43. For your best method, identify the top features that make each cluster distinctive.
44. Create one heatmap or table showing cluster feature profiles.
45. Explain at least three clusters or noise groups in plain language.
46. Connect each explanation to actual feature values from your notebook.

Cluster or Noise Group	Distinctive Features	Possible Interpretation	Evidence from Results

Required outputs:

- Cluster profile table
- Cluster profile heatmap or sorted feature summary
- Plain-language explanation of at least three clusters or noise groups
- Evidence-based interpretation using actual feature values

Part 9: Final Method Comparison and Recommendation

Create a final comparison and recommend which clustering method is most useful for this dataset.

47. Compare K-Means, hierarchical clustering, and DBSCAN in one table.
48. Discuss which method produced the clearest cluster structure.
49. Discuss which method was easiest to explain.
50. Discuss whether DBSCAN noise points were useful or mostly parameter artifacts.
51. Recommend one method for exploratory malware feature analysis and justify your answer.

Method	Main Strength	Main Weakness	Best Use in This Assignment
K-Means			
Hierarchical Clustering			
DBSCAN			

Required outputs:

- Final method comparison table
- Written recommendation paragraph
- Short discussion of limitations

Part 10: AI-Assisted Interpretation and Documentation

Use AI tools only as an assistant for interpretation, debugging, or clarity. You must document the prompts and verify the output. This part is worth 10 points.

Minimum requirement:

52. Use at least two meaningful AI prompts related to this assignment.
53. At least one prompt must focus on interpreting your actual cluster-profile or parameter-sweep results.
54. Include the exact prompt text in your notebook.
55. Summarize the AI output without copying a long answer verbatim.
56. Explain how you verified, corrected, rejected, or modified the AI output.
57. End with a clear AI Use Statement.

Suggested prompts:

- I ran K-Means on malware static and dynamic features. Here is my cluster profile table: [paste small aggregate table only]. Explain what patterns might separate the clusters, but do not invent facts not shown in the table.
- Here is my DBSCAN parameter sweep table with eps, MinPts, number of clusters, noise rate, and silhouette score. Which settings look reasonable and why?
- Explain the difference between single, complete, average, and Ward linkage using my clustering results as context.
- Review this paragraph for clarity. Do not change the meaning or add results I did not report: [paste your paragraph].

Required AI documentation:

58. Take Screenshots of your prompts and the output you received by executing such prompts.
59. Compile all of your screenshots under a docx file, name it: **Assignment3-LastName.docx**

Required AI Use Statement format:

AI Use Statement: I used [tool name] to help with [specific task]. I reviewed, tested, and modified the output before including it in my assignment.

AI Use Statement: I did not use AI tools for this assignment.

What to Submit

Submit one Jupyter Notebook file only:

1. Assignment3-LastName.ipynb
2. Assignment3-LastName.docx

The **notebook** must include code, outputs, plots, tables, written explanations, cluster interpretations, and the AI Use Statement.
AI prompts must be documented in the **docx** file.

Grading Rubric

Component	Points
Dataset loading, labeling, and combination	10
Data cleaning and feature selection	10
Scaling, exploratory analysis, and PCA visualization	10
K-Means clustering and K selection	10
Hierarchical clustering and linkage comparison	10
DBSCAN parameter analysis	10
Cluster evaluation and label comparison	10
Cluster profile analysis and interpretation	10
Final method comparison and recommendation	10
AI-assisted interpretation and documentation	10
Total	100

Important Notes

- Do not use source_label, binary_label, file names, hashes, or identifiers as clustering features.
- Do not report only one metric. Use silhouette score, cluster counts, noise rate, crosstabs, and cluster profiles.
- A higher silhouette score does not automatically mean the clusters have meaningful malware-analysis value.
- DBSCAN noise points are not automatically malicious. They are unusual under the selected parameter settings.
- Your explanations must be written in your own words.
- You must be able to explain your code, plots, clustering choices, and AI use documentation if asked.